MSBX 5415 – Advanced Data Analytics Final Project Report

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Linear Regression Modeling to Predict AirBnb Prices

Introduction and Analysis Goal

Our team's goal is to use Inside Airbnb data to develop interpretable, city-specific linear regression models for Austin and Zurich that quantify how property, listing and host attributes influence prices. Such models can help deliver reliable predictions for baseline pricing and inform a comparison of price impacting factors across markets to inform pricing strategy.

Through linear regression, backwards stepping, residual analysis, and data visualization, our team aims to test the effect of a combination of observable variables on the listing price on Airbnb in both cities. The use of both a national and international dataset will allow our team to compare how some variables may have different statistical significance to the price of a listing based on the country the listing is located in.

Data Source(s)

Datasets utilized in analysis were taken from the Inside Airbnb website. Inside Airbnb is a public data project which aims to provide "data and advocacy about Airbnb's impact on residential communities". All of the data posted by Inside Airbnb is public data gathered via scraping the official Airbnb website and the listings posted to it.

Dataset Summary

Two large datasets were used in analysis, each containing observations of Airbnb listings across their respective cities. The first dataset includes Airbnb listings in the city of Austin, Texas, while the second dataset includes Airbnb listings in the city of Zurich, Switzerland. These cities were chosen due to their similarities and differences. Both cities are tourist destinations and are home to large and growing work sectors. Austin, however, is a typical 'weekend destination', regularly hosting sporting events, festivals, and bachelor/bachelorette parties. These events are not as present in Zurich, but in contrast, Zurich hosts many professional conferences and extended work travel.

Each dataset contains 79 common variables. Variables include numeric fields such as the number of people the property accommodates, bedrooms, beds, bathrooms, the minimum and maximum nights allowed for a stay, and more. The dataset contains categorical variables such as room type (entire home/apt, private room, shared room, hotel), and logical variables, such as whether or not the host has a profile picture or identity verification (T/F).

Initial Data Cleaning and Preparation

Upon initial review of our dependent variable price, 4479 and 798 NA price values were found in the Austin and Zurich datasets, respectively. The rows containing these NA values were subsequently dropped from both datasets, to ensure accuracy when the linear regression model was created and run. Next, our team converted the price column to a numeric value, as it was initially shown as a character value due to the existence of dollar signs and commas within the values.

Rows that did not contain a price value were also missing information for other variables, so the removal of NA price rows significantly decreased the number of NA values in remaining rows. Our team checked for NA values in all the predictor variables that would be used in our initial linear regression model. An additional 34 rows were removed from the Austin dataset due to NA values remaining for the bathrooms, bedrooms, and beds variables. An additional 70 rows were removed from the Zurich dataset due to NA values remaining for the bathroom, beds, host profile picture, host identity verification, and host listings count variables. It is important to note that 59 rows were missing values for the same 3 variables: host profile picture, host identity verification, and host listings count. Given that, relative to the overall size of the dataset, a small number of rows containing NA values remained; these rows were dropped rather than attempting an imputation.

Next, boxplots were created of both datasets with the price variable as the x-axis. Multiple boxplots were generated with different maximum prices in order to easily identify any significant outliers (see all plots, Appendix A). Based on these boxplots, we determined that prices over $1250 and $875 were outliers and should be removed from the Austin and Zurich datasets, respectively. Once outliers were removed, histograms of the price variable were created for both datasets. Both histograms showed a right skew (see all plots, Appendix B), indicating that a log-transformation of the price variable would be necessary for linear regression analysis.

Following log-transformation of the price variable, cleaned versions of both datasets were created, with the following predictor variables included:

- host_listings_count (Numerical)
- host_has_profile_pic (Logical)
- host_identity_verified (Logical)
- room_type (Categorical)
- Accommodates (Numerical)
- Bathrooms (Numerical)
- Bedrooms (Numerical)
- Beds (Numerical)
- minimum_nights (Numerical)
- maximum_nights (Numerical)

With all outliers and NA values removed, the cleaned Austin dataset contained 10674 observations, and the cleaned Zurich dataset contained 2433 observations. The categorical and logical variables remaining in the cleaned datasets were then factored to later be used in the linear regression models. Additionally, the host_listings_count numerical variable was also factored into buckets to separate individual hosts with singular listings, smaller hosting groups, and larger property management groups. These listing buckets include: 1 listing, 2-10 listings, 10-30 listings, and 30+ listings.

Finally, the cleaned Austin and Zurich datasets were divided into training and testing groups using a 70-30 split, with 70% of the cleaned data from each set being used for training, and 30% of the cleaned data from each set being used for testing.

Initial Linear Regression Model and Interpretation:

A linear regression model was run on the log-transformed price variable as a function of the predictor variables remaining in the cleaned datasets (listed above).

This initial model suggests that the shared room type has the noticeably largest detrimental effect on listing prices in Austin, with this room category decreasing price by almost -180%. This was followed by private room, with this category decreasing price by almost -45%. Conversely, the number of bathrooms could lead to an increase in price by almost 23%, with the number of people accommodated following with an increase of about 5.5%. Room types also

presented a detrimental effect to price in Zurich, though this time private rooms led to the highest price decrease of around -40%. In terms of increase, the number of people accommodated, the number of bathrooms, and the number of bedrooms, also led to similar increases in price of around 11-12%. The initial Austin price model saw an adjusted R^2 value of 0.52, and the initial Zurich price model saw an adjusted R^2 value of 0.43

The results indicate that capacity and room type dominate pricing power in both markets. Meanwhile, host variables, such as host identity verification, had relatively little effect. The slightly lower adjusted R^2 value seen in the initial Zurich model suggests that a larger amount of pricing variation in that city may be explained unobserved variables than in Austin, where a slightly larger amount of pricing variation can be explained by the variables that Airbnb hosts are able to directly measure and post to their listings.

Final Linear Regression Model and Interpretation:

After the initial model was run, and both the significance and coefficients of each variable were noted, the final model was created via bidirectional stepping of the initial model. Bidirectional stepping iteratively adds and removes variables, eventually leading to a model that is optimized for both fit and complexity. The results of this stepwise selection showed the following variables remained in the final models for each city:

- **Common to both cities:** *room type*, *bathrooms*, *bedrooms*, *accommodates*, and *host listings count*
- **Austin only:** *minimum_nights*.
- **Zurich only:** *maximum_nights* and *beds*.

As previously noted, the linear regression was performed on a log-transformed price variable, meaning that the model coefficients need to be interpreted carefully. The most meaningful interpretation of the coefficients is a percent change per unit change in the predictor variable. The percent change was calculated by 'undoing' the log transformation via the following formula: exp(coefficient) - 1. Once transformed, the percent change can easily be interpreted. For example, in Austin our model predicts that for every additional bathroom that a property has, the price increases by 26.16%. For factored variables, the percent change is a change from the model's baseline. For example, in Austin, our model predicts that a shared room will be priced 36.04% lower than an entire home/apt with other features remaining the same.

An additional interpretation of the model results is an average dollar impact on the price. Using the median price of listings in each training set and the percent changes, an average dollar impact can be calculated (see all tables, Appendix C). For example, based on median listing price, our model predicts that in Austin, each additional bathroom will increase the listing price by $35.31, on average. This interpretation is useful for getting a quick, general understanding of the dollar impact of different variables, but should be used with caution, as it is an average dollar impact, based on the median price. The percent changes in price for each variable are the most robust and helpful outcome of these linear regression models.

Austin Final Model Interpretation

As expected, the model predicts that the listing price will increase as the number of bedrooms, bathrooms, and accommodatable people increases. The model predicts that the list price will increase by ~5.83% and 4.26% for each additional person and bedroom, respectively. The model reveals an interesting impact prediction for the number of bathrooms, with each additional bathroom increasing the predicted price by ~26.16%. The model predictions for room type also match expectation, with private rooms and shared rooms seeing a ~36.04% and ~83.41% price decrease, respectively, compared to an entire home/apt. The model also predicts that hotel rooms will be priced ~13.49% higher than entire homes/apts with the same features, however this change is the least statistically significant of room type predictors. The model predictions based on host listing count are perhaps the most interesting. From the baseline of a host with only 1 listing, hosts with 2-10 listings price prediction decrease by ~3.12%. Hosts with 10-30 listings and 30+ listings see a ~20.36% and ~9.77% price prediction increase, respectively. We suspect that host listing count does not tell the entire story, and that the property and host rating, host responsiveness, and other underlying factors would help inform why the model predicts these behaviors. Minimum number of nights is included in the model, but has the smallest impact of ~ -.21%. Finally, the model predicts a ~4.43% decrease in price when the host's identity is verified. On first glance, this may be counter-intuitive, but we suspect that confounding effects and/or a few highly priced listings with unverified hosts led to this prediction.

Zurich Final Model Interpretation

Similar to the Austin model, the Zurich model predicts that the listing price will increase as the number of bedrooms, bathrooms, and accommodatable people increases. The Zurich

model, however, predicts that these variables have relatively similar impact on price, as the model predicts a ~13.61%, ~13.51% and ~12.56% increase in price per unit increase in accommodatable people, bathrooms, and bedrooms, respectively. The model predictions for room type also match expectation, with private rooms and shared rooms seeing a ~33.34% and ~4.41% price decrease, respectively, compared to an entire home/apt. The model also predicts that hotel rooms will be priced ~38.66% higher than entire homes/apts with the same features, however this change is the least statistically significant of room type predictors. The model predictions based on host listing count in Zurich are perhaps more intuitive than the results for Austin. From the baseline of a host with only 1 listing, hosts with 2-10 listings price prediction decrease by ~3.83%, hosts with 10-30 listings decrease by ~ 5.93% and hosts with 30+ listings result in a ~ 8.23% decrease in predicted price. This trend aligns with a logical thought that sole-proprietor hosts will charge more (higher cleaning and opportunity costs) than a property management company (lower cleaning and opportunity costs, lower necessary margins). The Zurich model predicts a decrease in predicted price by ~4.73% for each additional bed. Overlapping effects with the accommodates and bedrooms variable is a likely explanation for this. Holding 'accommodates' and bedrooms constant, additional beds on a property may indicate lower quality or 'cramming'. Similarly, after a certain point, most additional beds may be sofa beds or air mattresses, which likely Maximum number of nights is included in the model but has the smallest impact of ~ 0.03%.

Model Performance

Prior to using the models for prediction, the residuals of each model were examined. The models for both Austin and Zurich have relatively normally distributed residuals (see all plots, Appendix D). Both residual lots have a slight 'edge' on the upper right corner, likely caused by dependent variable truncation (removing high price outliers).

The final models for Austin and Zurich both have adjusted R^2 values similar to the initial models. R^2 values near 0.5 (0.5247 and 0.4345 for Austin and Zurich, respectively) do not necessarily mean that the models are poor or irrelevant. In this context, this simply means that ~50% of the variability in price can be explained by our model. Thus, ~50% of the variability in price is explained by other factors. Plotting the model predicted price vs the actual

price, and comparing the scatterplot to a 'perfect prediction' line provides a visual for the conclusion that some variability in price is not explained by our model (see all plots, Appendix E).

After inspecting and interpreting each model, the models were used to predict the price of listings in the test datasets, and then compared to the actual test dataset prices for accuracy. Full prediction accuracy results can be found in Appendix F. Standalone accuracy results are difficult to interpret. To provide context for the prediction accuracy, also calculated the accuracy of models that predicts the mean and median price of the training set for every listing (see all tables, Appendix F). Our final models have lower root mean squared error and mean absolute error, and thus, higher accuracy, when compared to the mean and median models. These accuracy measures could be further improved with the inclusion of other relevant variables.

Conclusion

The regression analysis of both the Austin and Zurich Airbnb datasets provides insight into and allows for comparison of the elements of listing prices across two unique markets on the national and international level. Results indicate that factors including room type, number of bedrooms and bathrooms, number of people accommodated, and host listing count influence price in both markets. However, each city has its own unique listing aspect, with prices in Austin seeing impact from the minimum of nights required, while Zurich prices are differentiated by the inverse maximum number of nights, as well as the number of beds.

However, these models account for only about 50% of the variation in the prices of Airbnb listings. This suggests that a multitude of unobserved variables, such as ambience, perceived furniture and amenity quality, proximity to important locations, neighborhood/safety, and other less easily measured categories account for the remaining 50%, and that there is therefore likely some extent of host discretion in setting Airbnb prices. When determining a competitive listing price, hosts should be certain to include both of these sets of variables in their thought process.

Sources Cited

Inside Airbnb. (2020, July 27). *Detailed listings data — Austin (listings.csv.gz)* [Data set]. Inside Airbnb. https://insideairbnb.com/get-the-data/

Inside Airbnb. (2020, July 27). *Detailed listings data — Zurich (listings.csv.gz)* [Data set]. Inside Airbnb. https://insideairbnb.com/get-the-data/

Appendices

Appendix A – Price Boxplots

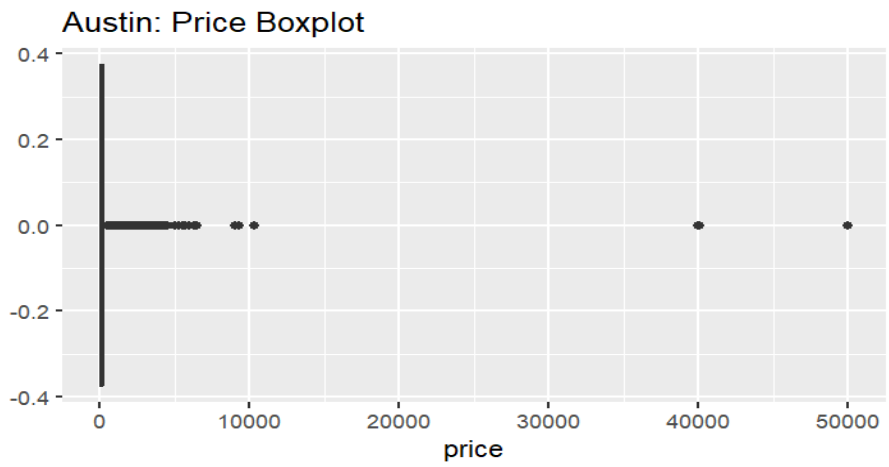

Figure A1 – Austin Price Boxplot (Full)

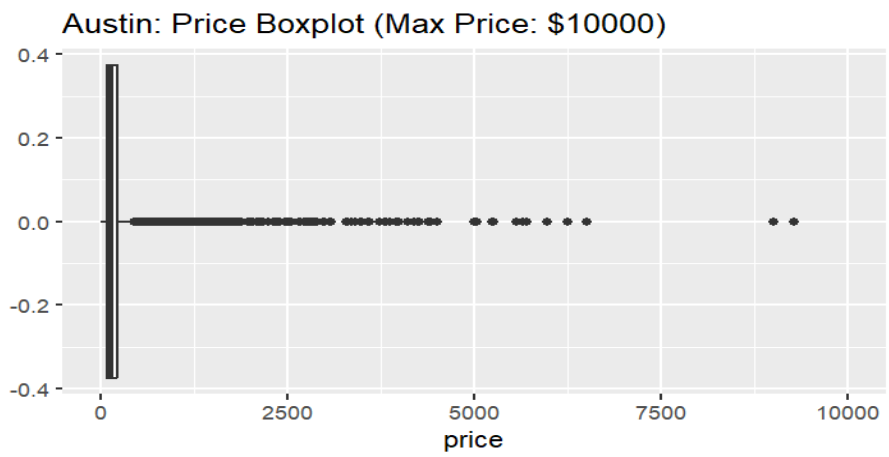

Figure A2 – Austin Price Boxplot (Max Price: $10000)

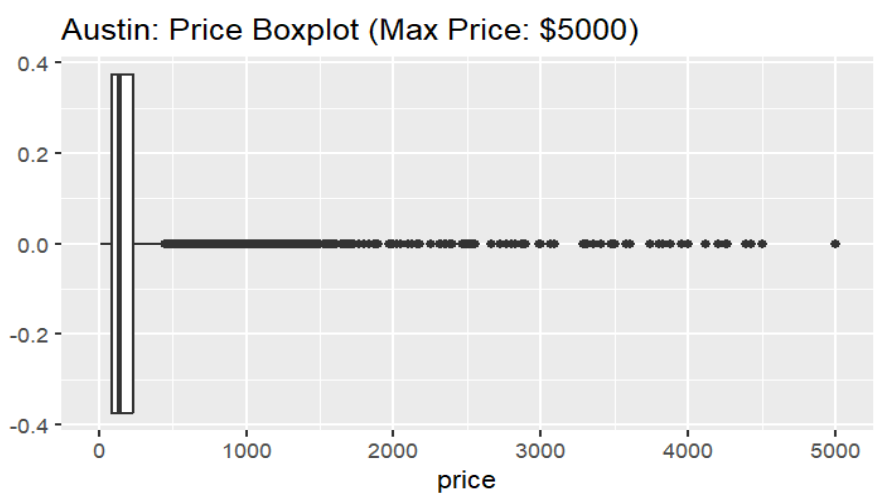

Figure A3 – Austin Price Boxplot (Max Price: $5000)

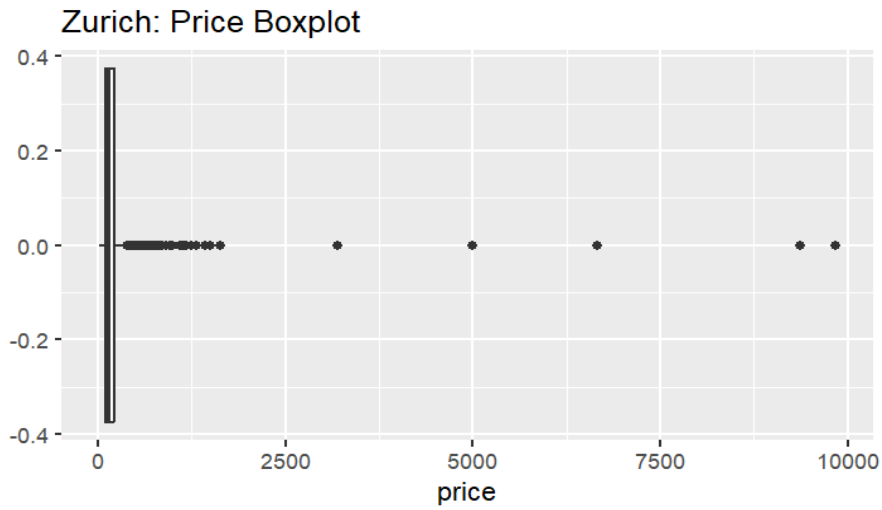

Figure A4 – Zurich Price Boxplot (Full)

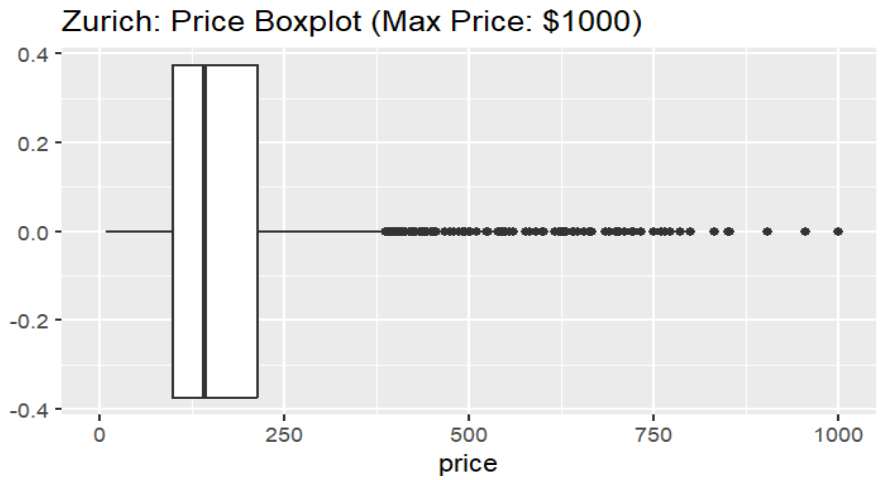

Figure A5 – Zurich Price Boxplot (Max Price: $1000)

Appendix B – Price Histograms

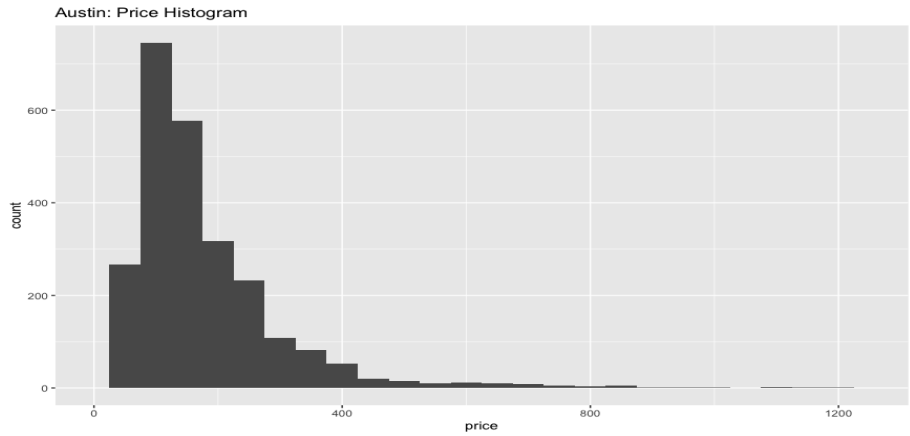

Figure B1 – Austin Price Histogram

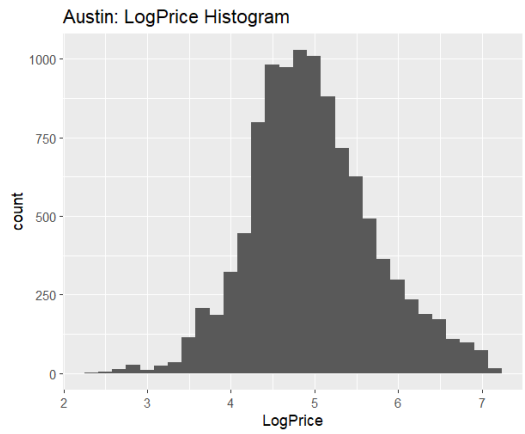

Figure B2 – Austin LogPrice Histogram

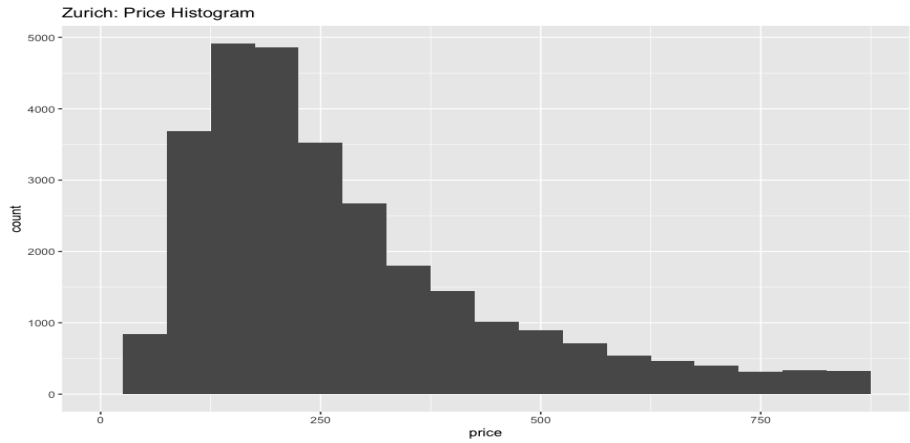

Figure B3 – Zurich Price Histogram

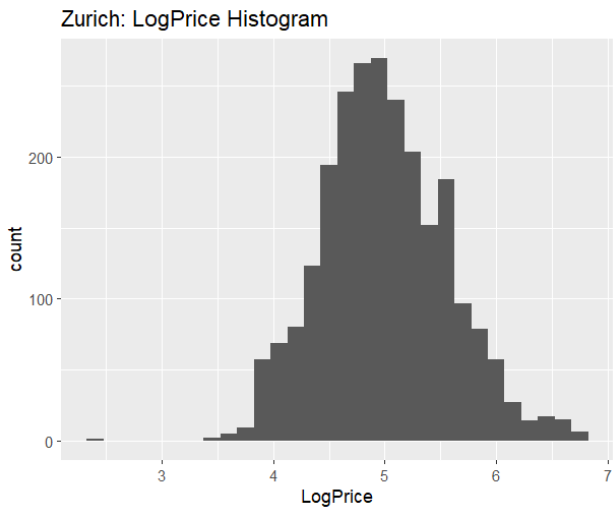

Figure B4 – Zurich LogPrice Histogram

Appendix C – Model Results

Table C1 – Austin Model Results

Variable	Coefficient (log)	% Change	Est. $ Impact
(Intercept)	4.313711020	NA	NA
host_listings_count.L	-0.031715246	-3.12 %	$ -4.21
host_listings_count.Q	0.185295045	20.36 %	$ 27.48
host_listings_count.C	0.093208642	9.77 %	$ 13.19
host_identity_verifiedTRUE	-0.045327420	-4.43 %	$ -5.98
room_typeHotel room	0.126561024	13.49 %	$ 18.21
room_typePrivate room	-0.446861021	-36.04 %	$ -48.65
room_typeShared room	-1.796570089	-83.41 %	$ -112.61

accommodates	0.056657488	5.83 %	$ 7.87
bathrooms	0.232355473	26.16 %	$ 35.31
bedrooms	0.041698665	4.26 %	$ 5.75
minimum_nights	-0.002103609	-0.21 %	$ -0.28

Table C2 – Zurich Model Results

Variable	Coefficient (log)	% Change	Est. $ Impact
(Intercept)	4.4104765131	NA	NA
host_listings_count.L	-0.0390897659	-3.83 %	$ -5.52
host_listings_count.Q	-0.0611230569	-5.93 %	$ -8.54
host_listings_count.C	-0.0858525822	-8.23 %	$ -11.85
room_typeHotel room	0.3268231111	38.66 %	$ 55.66
room_typePrivate room	-0.4055870691	-33.34 %	$ -48.01
room_typeShared room	-0.0451892851	-4.42 %	$ -6.36
accommodates	0.1275972772	13.61 %	$ 19.6
bathrooms	0.1267008831	13.51 %	$ 19.45
bedrooms	0.1183403094	12.56 %	$ 18.09
beds	-0.0484236628	-4.73 %	$ -6.81
maximum_nights	0.0002595593	0.03 %	$ 0.04

Appendix D – Model Residual Plots

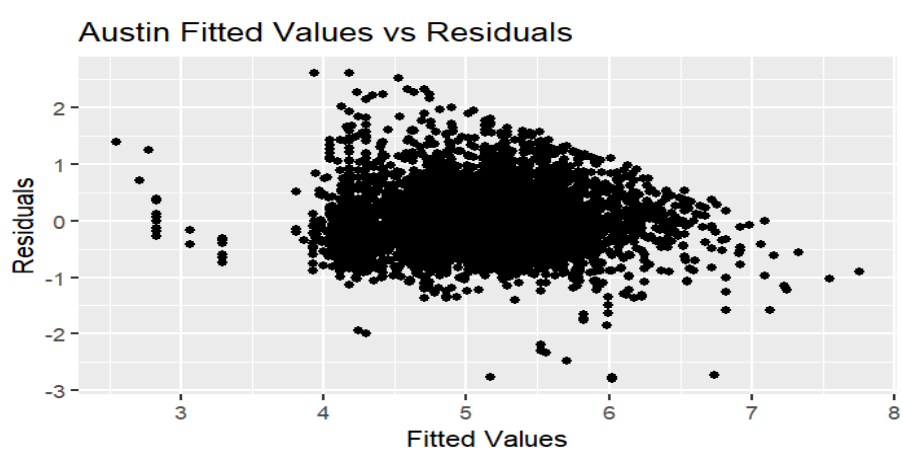

Figure D1 – Austin Final Model Residuals Scatterplot

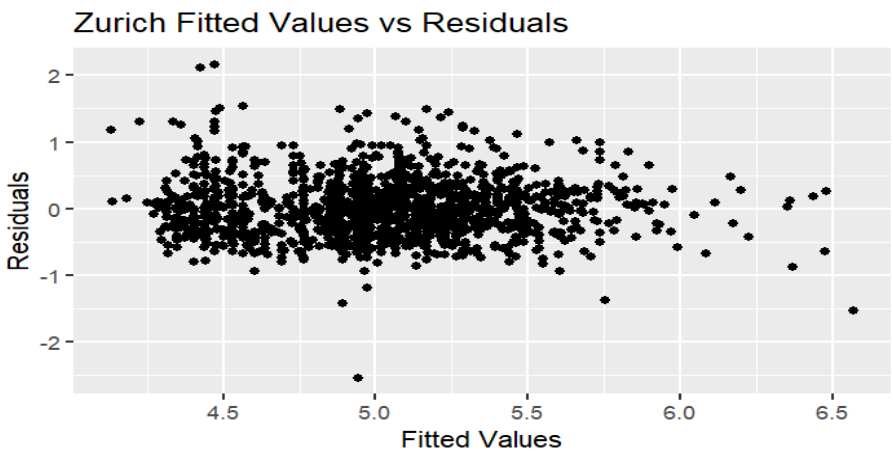

Figure D2 – Zurich Final Model Residuals Scatterplot

Appendix E – Predicted vs Actual Price Plots

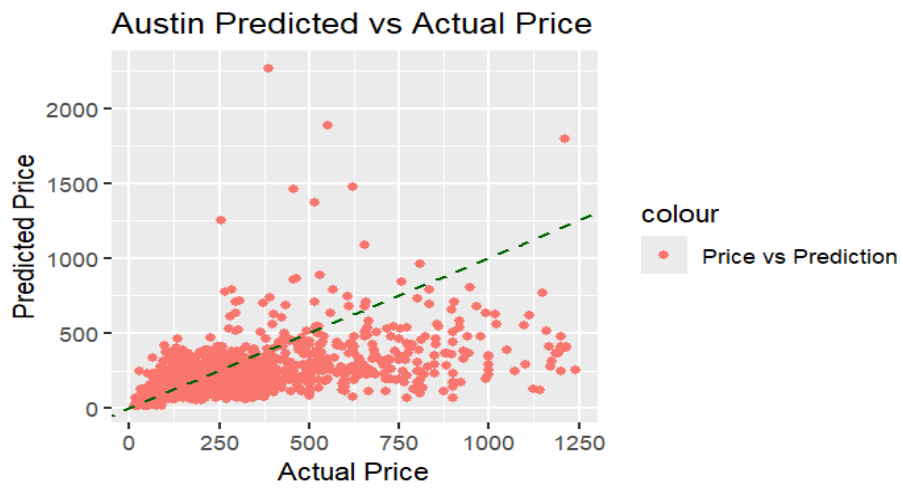

Figure E1 – Austin Predicted vs Actual Prices

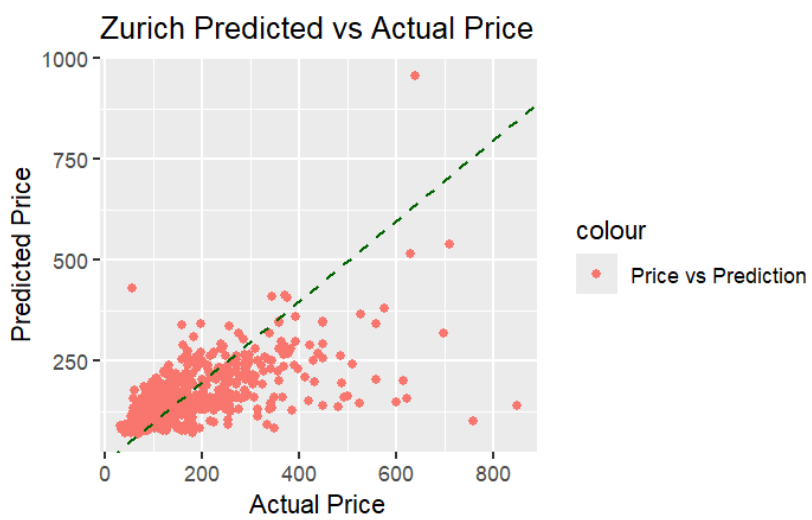

Figure E2 – Zurich Predicted vs Actual Price

Appendix F – Model Accuracy Results

Table F1 – Accuracy Results for Final Models

City	ME ($)	RMSE ($)	MAE ($)	MPE (%)	MAPE (%)
Austin	28.44	156.99	82.29	-10.77	39.72
Zurich	13.43	90.17	55.93	-10.84	34.64

Table F2- Accuracy Results for Mean and Median Price Models

City	Baseline	ME ($)	RMSE ($)	MAE ($)	MPE (%)	MAPE(%)
Austin	Median	63.43	198.53	111.53	-20.74	62.75
Austin	Mean	6.61	188.24	122.61	-71.56	96.50
Zurich	Median	26.52	116.19	75.84	-17.31	49.38
Zurich	Mean	-4.40	113.21	82.01	-42.50	64.07